

**HAEJU, Korea, Democratic People's Republic of**WMO: **470690**Lat: **38.033N**Lon: **125.700E**Elev: **81**StdP: **100.36**Time Zone: **9.00 (E09)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -11.1      | -9.3 | -20.0                         | 0.6 | -7.7 | -18.0 | 0.8 | -7.4 | 11.2                  | -4.9 | 9.6 | -3.8 | 3.3                   | 320  | 0.393 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 5.7                    | 31.4            | 25.8 | 30.0 | 24.7 | 28.7 | 24.1 | 26.8                | 30.0 | 26.1 | 28.8 | 25.3 | 27.6 | 2.4                  | 230  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 25.9                            | 21.5 | 28.8 | 25.2 | 20.6 | 27.8 | 24.6 | 19.8 | 27.1 | 85.0          | 30.2 | 81.5 | 28.8 | 78.4 | 27.7 | 30.1              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |            |            | Extreme Annual Temperature |             |                    |            | n-Year Return Period Values of Extreme Temperature |             |              |             |              |             |              |             |  |  |
|-------------------|------------|------------|----------------------------|-------------|--------------------|------------|----------------------------------------------------|-------------|--------------|-------------|--------------|-------------|--------------|-------------|--|--|
|                   |            |            | Mean                       |             | Standard Deviation |            | n=5 years                                          |             | n=10 years   |             | n=20 years   |             | n=50 years   |             |  |  |
| 1%                | 2.5%       | 5%         | Min                        | Max         | Min                | Max        | Min                                                | Max         | Min          | Max         | Min          | Max         | Min          | Max         |  |  |
| (a)               | (b)        | (c)        | (e)                        | (f)         | (g)                | (h)        | (i)                                                | (j)         | (k)          | (l)         | (m)          | (n)         | (o)          | (p)         |  |  |
| (4) <b>9.5</b>    | <b>8.1</b> | <b>6.8</b> | <b>-13.5</b>               | <b>33.1</b> | <b>2.5</b>         | <b>1.2</b> | <b>-15.4</b>                                       | <b>34.0</b> | <b>-16.8</b> | <b>34.7</b> | <b>-18.3</b> | <b>35.3</b> | <b>-20.1</b> | <b>36.2</b> |  |  |
| (5)               |            |            | <b>-14.6</b>               | <b>28.0</b> | <b>2.3</b>         | <b>1.1</b> | <b>-16.3</b>                                       | <b>28.8</b> | <b>-17.6</b> | <b>29.4</b> | <b>-18.9</b> | <b>30.0</b> | <b>-20.6</b> | <b>30.8</b> |  |  |

**Monthly Climatic Design Conditions**

|                                                                               |             | Annual       | Jan          | Feb          | Mar          | Apr          | May          | Jun          | Jul          | Aug          | Sep          | Oct          | Nov          | Dec         |
|-------------------------------------------------------------------------------|-------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|-------------|
|                                                                               |             | (d)          | (e)          | (f)          | (g)          | (h)          | (i)          | (j)          | (k)          | (l)          | (m)          | (n)          | (o)          | (p)         |
| (6) <b>Temperatures, Degree-Days and Degree-Hours</b>                         | DBAvg       | <b>12.2</b>  | <b>-2.7</b>  | <b>-0.2</b>  | <b>4.9</b>   | <b>11.4</b>  | <b>16.9</b>  | <b>21.6</b>  | <b>24.4</b>  | <b>25.6</b>  | <b>21.4</b>  | <b>15.1</b>  | <b>6.9</b>   | <b>-0.1</b> |
|                                                                               | DBStd       | <b>10.26</b> | <b>3.82</b>  | <b>3.67</b>  | <b>3.25</b>  | <b>3.30</b>  | <b>2.75</b>  | <b>2.20</b>  | <b>2.10</b>  | <b>2.26</b>  | <b>2.39</b>  | <b>3.29</b>  | <b>4.41</b>  | <b>4.00</b> |
|                                                                               | HDD10.0     | <b>1289</b>  | <b>393</b>   | <b>286</b>   | <b>160</b>   | <b>21</b>    | <b>0</b>     | <b>0</b>     | <b>0</b>     | <b>0</b>     | <b>4</b>     | <b>112</b>   | <b>313</b>   |             |
|                                                                               | HDD18.3     | <b>2884</b>  | <b>652</b>   | <b>519</b>   | <b>416</b>   | <b>208</b>   | <b>62</b>    | <b>3</b>     | <b>0</b>     | <b>0</b>     | <b>4</b>     | <b>107</b>   | <b>343</b>   | <b>572</b>  |
|                                                                               | CDD10.0     | <b>2079</b>  | <b>0</b>     | <b>0</b>     | <b>2</b>     | <b>64</b>    | <b>214</b>   | <b>347</b>   | <b>447</b>   | <b>483</b>   | <b>341</b>   | <b>163</b>   | <b>18</b>    | <b>0</b>    |
|                                                                               | CDD18.3     | <b>633</b>   | <b>0</b>     | <b>0</b>     | <b>0</b>     | <b>1</b>     | <b>17</b>    | <b>99</b>    | <b>188</b>   | <b>226</b>   | <b>96</b>    | <b>7</b>     | <b>0</b>     | <b>0</b>    |
|                                                                               | CDH23.3     | <b>3953</b>  | <b>0</b>     | <b>0</b>     | <b>0</b>     | <b>3</b>     | <b>91</b>    | <b>441</b>   | <b>1143</b>  | <b>1871</b>  | <b>390</b>   | <b>16</b>    | <b>0</b>     | <b>0</b>    |
|                                                                               | CDH26.7     | <b>910</b>   | <b>0</b>     | <b>0</b>     | <b>0</b>     | <b>0</b>     | <b>9</b>     | <b>78</b>    | <b>256</b>   | <b>524</b>   | <b>43</b>    | <b>0</b>     | <b>0</b>     | <b>0</b>    |
| (14) <b>Wind</b>                                                              | WSAvg       | <b>2.3</b>   | <b>2.3</b>   | <b>2.5</b>   | <b>2.7</b>   | <b>2.7</b>   | <b>2.4</b>   | <b>2.0</b>   | <b>2.1</b>   | <b>2.2</b>   | <b>1.9</b>   | <b>2.0</b>   | <b>2.2</b>   | <b>2.3</b>  |
| (15) <b>Precipitation</b>                                                     | PrecAvg     | <b>1064</b>  | <b>10</b>    | <b>18</b>    | <b>25</b>    | <b>52</b>    | <b>102</b>   | <b>118</b>   | <b>320</b>   | <b>233</b>   | <b>109</b>   | <b>42</b>    | <b>37</b>    | <b>15</b>   |
|                                                                               | PrecMax     | <b>2053</b>  | <b>30</b>    | <b>59</b>    | <b>89</b>    | <b>122</b>   | <b>235</b>   | <b>463</b>   | <b>739</b>   | <b>501</b>   | <b>463</b>   | <b>141</b>   | <b>89</b>    | <b>44</b>   |
|                                                                               | PrecMin     | <b>502</b>   | <b>1</b>     | <b>0</b>     | <b>0</b>     | <b>0</b>     | <b>8</b>     | <b>16</b>    | <b>97</b>    | <b>39</b>    | <b>13</b>    | <b>3</b>     | <b>5</b>     | <b>2</b>    |
|                                                                               | PrecStd     | <b>314</b>   | <b>8</b>     | <b>15</b>    | <b>20</b>    | <b>29</b>    | <b>61</b>    | <b>88</b>    | <b>156</b>   | <b>126</b>   | <b>109</b>   | <b>38</b>    | <b>23</b>    | <b>9</b>    |
| (19) <b>Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures</b> | 0.4%        | DB           | <b>8.3</b>   | <b>11.3</b>  | <b>16.3</b>  | <b>23.4</b>  | <b>27.9</b>  | <b>30.8</b>  | <b>33.0</b>  | <b>33.5</b>  | <b>29.5</b>  | <b>25.4</b>  | <b>18.5</b>  | <b>10.7</b> |
|                                                                               |             | MCWB         | <b>5.9</b>   | <b>7.7</b>   | <b>10.3</b>  | <b>15.9</b>  | <b>18.8</b>  | <b>22.1</b>  | <b>26.3</b>  | <b>27.4</b>  | <b>23.2</b>  | <b>18.9</b>  | <b>13.9</b>  | <b>7.8</b>  |
|                                                                               | 2%          | DB           | <b>5.6</b>   | <b>8.6</b>   | <b>13.4</b>  | <b>20.7</b>  | <b>25.7</b>  | <b>28.8</b>  | <b>30.8</b>  | <b>32.0</b>  | <b>27.9</b>  | <b>23.0</b>  | <b>16.2</b>  | <b>8.7</b>  |
|                                                                               |             | MCWB         | <b>3.1</b>   | <b>4.8</b>   | <b>8.3</b>   | <b>13.5</b>  | <b>17.7</b>  | <b>21.3</b>  | <b>25.6</b>  | <b>26.8</b>  | <b>22.4</b>  | <b>17.8</b>  | <b>13.3</b>  | <b>6.3</b>  |
|                                                                               | 5%          | DB           | <b>4.1</b>   | <b>6.6</b>   | <b>11.6</b>  | <b>18.6</b>  | <b>23.9</b>  | <b>27.2</b>  | <b>29.1</b>  | <b>30.6</b>  | <b>26.7</b>  | <b>21.6</b>  | <b>14.7</b>  | <b>7.2</b>  |
|                                                                               |             | MCWB         | <b>1.9</b>   | <b>3.5</b>   | <b>7.2</b>   | <b>12.3</b>  | <b>17.1</b>  | <b>20.9</b>  | <b>24.8</b>  | <b>25.6</b>  | <b>21.5</b>  | <b>16.8</b>  | <b>12.1</b>  | <b>5.0</b>  |
|                                                                               | 10%         | DB           | <b>2.7</b>   | <b>5.0</b>   | <b>10.0</b>  | <b>16.9</b>  | <b>22.1</b>  | <b>25.8</b>  | <b>27.9</b>  | <b>29.4</b>  | <b>25.6</b>  | <b>20.2</b>  | <b>13.2</b>  | <b>5.8</b>  |
|                                                                               |             | MCWB         | <b>0.7</b>   | <b>2.4</b>   | <b>6.4</b>   | <b>11.6</b>  | <b>16.3</b>  | <b>20.5</b>  | <b>24.3</b>  | <b>25.0</b>  | <b>20.7</b>  | <b>15.9</b>  | <b>10.5</b>  | <b>3.6</b>  |
| (27) <b>Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures</b> | 0.4%        | WB           | <b>6.1</b>   | <b>8.8</b>   | <b>11.7</b>  | <b>17.6</b>  | <b>21.2</b>  | <b>24.5</b>  | <b>27.6</b>  | <b>28.4</b>  | <b>25.0</b>  | <b>20.5</b>  | <b>16.1</b>  | <b>8.7</b>  |
|                                                                               |             | MCDB         | <b>7.7</b>   | <b>10.2</b>  | <b>14.3</b>  | <b>21.9</b>  | <b>25.4</b>  | <b>28.3</b>  | <b>30.9</b>  | <b>32.3</b>  | <b>27.6</b>  | <b>23.6</b>  | <b>17.0</b>  | <b>10.5</b> |
|                                                                               | 2%          | WB           | <b>3.8</b>   | <b>5.5</b>   | <b>9.5</b>   | <b>14.9</b>  | <b>19.3</b>  | <b>22.9</b>  | <b>26.4</b>  | <b>27.2</b>  | <b>23.8</b>  | <b>18.9</b>  | <b>14.1</b>  | <b>6.9</b>  |
|                                                                               |             | MCDB         | <b>5.2</b>   | <b>7.6</b>   | <b>12.1</b>  | <b>18.2</b>  | <b>23.8</b>  | <b>26.6</b>  | <b>29.6</b>  | <b>30.7</b>  | <b>26.3</b>  | <b>21.8</b>  | <b>15.4</b>  | <b>8.3</b>  |
|                                                                               | 5%          | WB           | <b>2.2</b>   | <b>4.0</b>   | <b>7.9</b>   | <b>13.6</b>  | <b>18.2</b>  | <b>22.1</b>  | <b>25.5</b>  | <b>26.5</b>  | <b>22.8</b>  | <b>17.8</b>  | <b>12.5</b>  | <b>5.3</b>  |
|                                                                               |             | MCDB         | <b>3.8</b>   | <b>5.9</b>   | <b>10.6</b>  | <b>17.3</b>  | <b>22.2</b>  | <b>25.4</b>  | <b>28.1</b>  | <b>29.5</b>  | <b>25.5</b>  | <b>20.4</b>  | <b>14.3</b>  | <b>7.0</b>  |
|                                                                               | 10%         | WB           | <b>0.9</b>   | <b>2.8</b>   | <b>6.8</b>   | <b>12.4</b>  | <b>17.2</b>  | <b>21.4</b>  | <b>24.8</b>  | <b>25.9</b>  | <b>21.8</b>  | <b>16.7</b>  | <b>10.8</b>  | <b>3.8</b>  |
|                                                                               |             | MCDB         | <b>2.5</b>   | <b>4.7</b>   | <b>9.5</b>   | <b>16.0</b>  | <b>20.6</b>  | <b>24.4</b>  | <b>26.9</b>  | <b>28.4</b>  | <b>24.6</b>  | <b>19.4</b>  | <b>13.0</b>  | <b>5.6</b>  |
| (35) <b>Mean Daily Temperature Range</b>                                      | 5% DB       | MDBR         | <b>6.7</b>   | <b>7.1</b>   | <b>7.6</b>   | <b>7.9</b>   | <b>7.9</b>   | <b>6.7</b>   | <b>4.7</b>   | <b>5.7</b>   | <b>7.2</b>   | <b>8.0</b>   | <b>6.9</b>   | <b>6.4</b>  |
|                                                                               |             | MCDBR        | <b>7.2</b>   | <b>8.2</b>   | <b>9.4</b>   | <b>10.1</b>  | <b>10.8</b>  | <b>9.5</b>   | <b>7.2</b>   | <b>7.2</b>   | <b>8.4</b>   | <b>9.1</b>   | <b>7.9</b>   | <b>7.3</b>  |
|                                                                               |             | MCWBR        | <b>5.8</b>   | <b>6.3</b>   | <b>6.3</b>   | <b>5.8</b>   | <b>5.9</b>   | <b>5.2</b>   | <b>4.3</b>   | <b>3.9</b>   | <b>4.8</b>   | <b>5.6</b>   | <b>6.0</b>   | <b>6.2</b>  |
|                                                                               |             | MCDBR        | <b>6.7</b>   | <b>7.3</b>   | <b>8.2</b>   | <b>8.4</b>   | <b>8.8</b>   | <b>7.2</b>   | <b>5.8</b>   | <b>6.1</b>   | <b>7.1</b>   | <b>7.8</b>   | <b>6.9</b>   | <b>6.9</b>  |
|                                                                               | 5% WB       | MCWBR        | <b>6.0</b>   | <b>6.3</b>   | <b>6.5</b>   | <b>5.7</b>   | <b>5.8</b>   | <b>4.6</b>   | <b>3.9</b>   | <b>3.6</b>   | <b>4.7</b>   | <b>5.7</b>   | <b>6.1</b>   | <b>6.4</b>  |
| (40) <b>Clear-Sky Solar Irradiance</b>                                        | taub        | <b>0.425</b> | <b>0.485</b> | <b>0.553</b> | <b>0.607</b> | <b>0.621</b> | <b>0.701</b> | <b>0.671</b> | <b>0.578</b> | <b>0.510</b> | <b>0.514</b> | <b>0.492</b> | <b>0.432</b> |             |
|                                                                               | taud        | <b>1.935</b> | <b>1.830</b> | <b>1.720</b> | <b>1.656</b> | <b>1.676</b> | <b>1.587</b> | <b>1.697</b> | <b>1.905</b> | <b>1.997</b> | <b>1.870</b> | <b>1.867</b> | <b>1.940</b> |             |
|                                                                               | Ebn at Noon | <b>701</b>   | <b>710</b>   | <b>705</b>   | <b>699</b>   | <b>700</b>   | <b>645</b>   | <b>660</b>   | <b>715</b>   | <b>738</b>   | <b>674</b>   | <b>623</b>   | <b>656</b>   |             |
|                                                                               | Edh at Noon | <b>139</b>   | <b>176</b>   | <b>217</b>   | <b>244</b>   | <b>244</b>   | <b>266</b>   | <b>237</b>   | <b>188</b>   | <b>162</b>   | <b>167</b>   | <b>149</b>   | <b>130</b>   |             |
| (44) <b>All-Sky Solar Radiation</b>                                           | RadAvg      | <b>2.40</b>  | <b>3.21</b>  | <b>4.34</b>  | <b>5.03</b>  | <b>5.60</b>  | <b>5.17</b>  | <b>4.16</b>  | <b>4.48</b>  | <b>4.33</b>  | <b>3.59</b>  | <b>2.38</b>  | <b>2.07</b>  |             |
|                                                                               | RadStd      | <b>0.17</b>  | <b>0.23</b>  | <b>0.36</b>  | <b>0.34</b>  | <b>0.52</b>  | <b>0.56</b>  | <b>0.59</b>  | <b>0.58</b>  | <b>0.40</b>  | <b>0.24</b>  | <b>0.24</b>  | <b>0.13</b>  |             |

**Historical Trends**

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| Regional (30 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
|                         | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +92     | +52     |